

REDROB AI CHALLENGE

Intelligent Candidate Discovery & Ranking

Ranking candidates the way a great recruiter would —
by understanding fit, not matching keywords.

100,000 candidates

• **Transparent multi-signal scoring**

• **CPU-only · offline · < 5 min**

Approach & Results

Great candidates hide in plain sight

Recruiters skim hundreds of profiles and still miss the right person — not because the talent isn't there, but because filters can't see what matters.

- **Keyword filters can't tell real from fake**

An "HR Manager" who lists LangChain, RAG and embeddings is not an AI engineer.

- **The best fit often has the wrong title**

A backend engineer who quietly shipped a recommendation system is a hidden gem.

- **Looking good on paper isn't being hireable**

A perfect profile that's been inactive for months won't actually take the call.

Our job: read the role, weigh the whole picture, and return a shortlist a recruiter can trust.

A transparent rubric, not a black box

- **Understand the role**

Parse the JD into a structured rubric — must-haves, anti-titles, disqualifiers, location.

- **Score the whole picture**

Six decomposable signals per candidate, each independently explainable.

- **Temper with reality**

Soft multipliers for disqualifiers and genuine availability.

- **Defend every pick**

Honeypots sink; the top 100 ships with grounded reasoning.



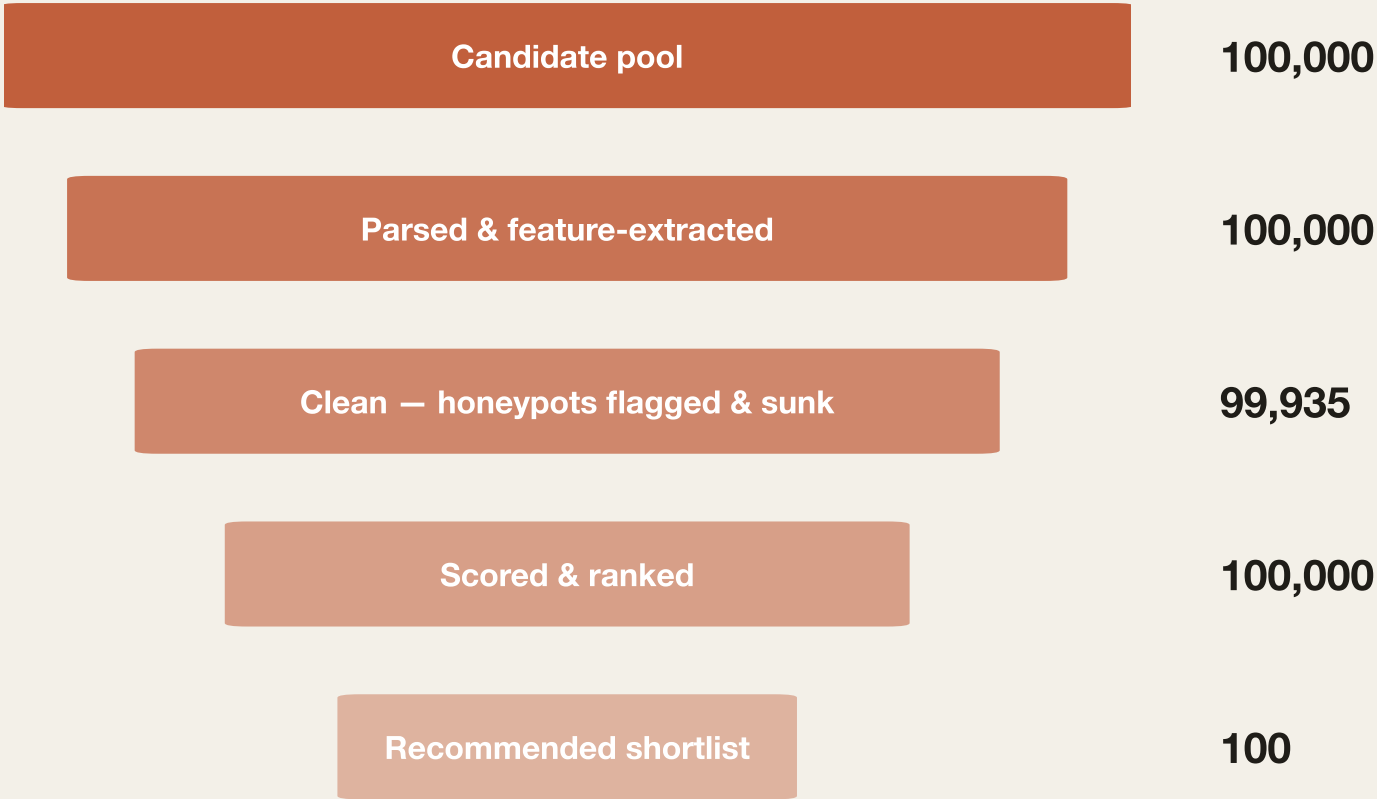
The pipeline, end to end

One streaming pass

65 honeypots flagged by contradiction checks — none reach the top 100.

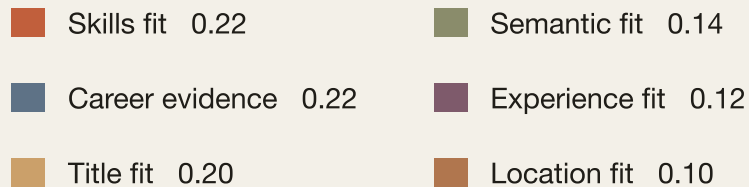
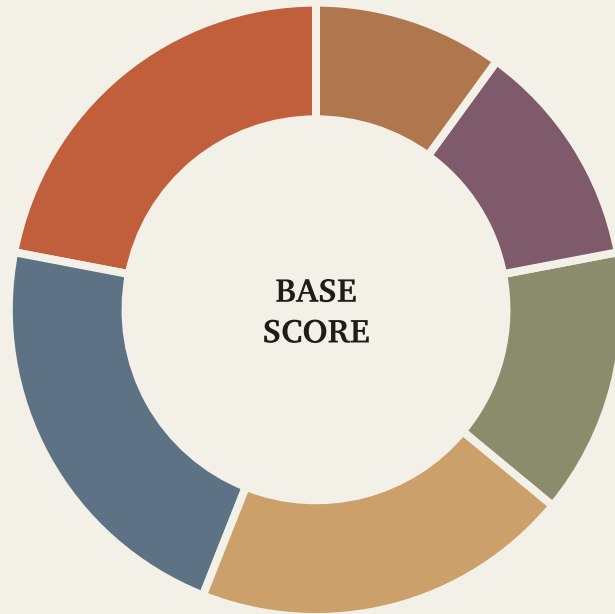
Constant memory

Compact records in RAM; no full-text bloat.



Funnel widths are schematic; the counts on the right are exact from the real run.

Six signals, each one explainable



— Skills fit

Trust-weighted family coverage. Trust = proficiency × duration × endorsements × Redrob's verified assessment — "expert, 0 months" ≈ 0.

— Career evidence

Free-text proof of shipping ranking / search / rec systems — the JD's #1 positive and the "hidden gem" signal.

— Title · Semantic · Fit

On-target role (junior / research discounted), MiniLM profile-vs-JD similarity, the 5–9y band and location.

Every component is logged per candidate — so the reasoning is grounded, never invented.

Built to beat the traps

65

honeypots flagged
by contradiction checks

0

honeypots in the
final top 100

3 / 3

known keyword-stuffer
traps excluded

— Title vs skills mismatch

An HR / Sales / Marketing title claiming AI skills is gated, not rewarded.

— Internal contradictions

A skill "used" longer than the whole career, or impossible role tenure.

— Self-claim ≠ truth

Platform-verified assessments discount inflated proficiencies directly.

— Research without production

The JD's explicit disqualifier — penalised, not credited.

CAND_0004989 · CAND_0001195 · CAND_0000339 — all kept out of the top 100.

Behavioral signals = real hireability

A flawless profile that ignores recruiters is, for hiring purposes, not available.

Behavior never invents fit — it modulates a candidate who already earns it on substance.

— Availability

Recruiter response rate, last-active recency, open-to-work, notice period
(clamped 0.40–1.05).

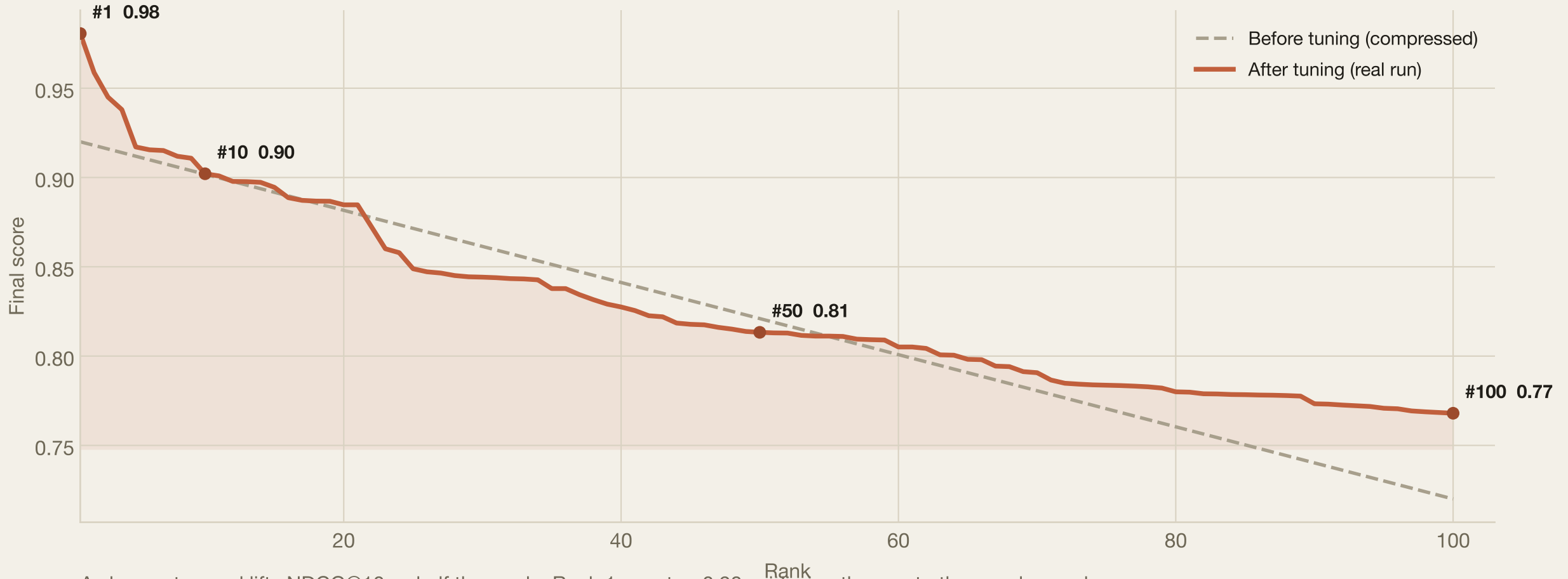
— Market interest

Recruiter-saves and search-appearances, scored against the pool's own
median / p90 — not a hardcoded threshold.

— Engagement & intent

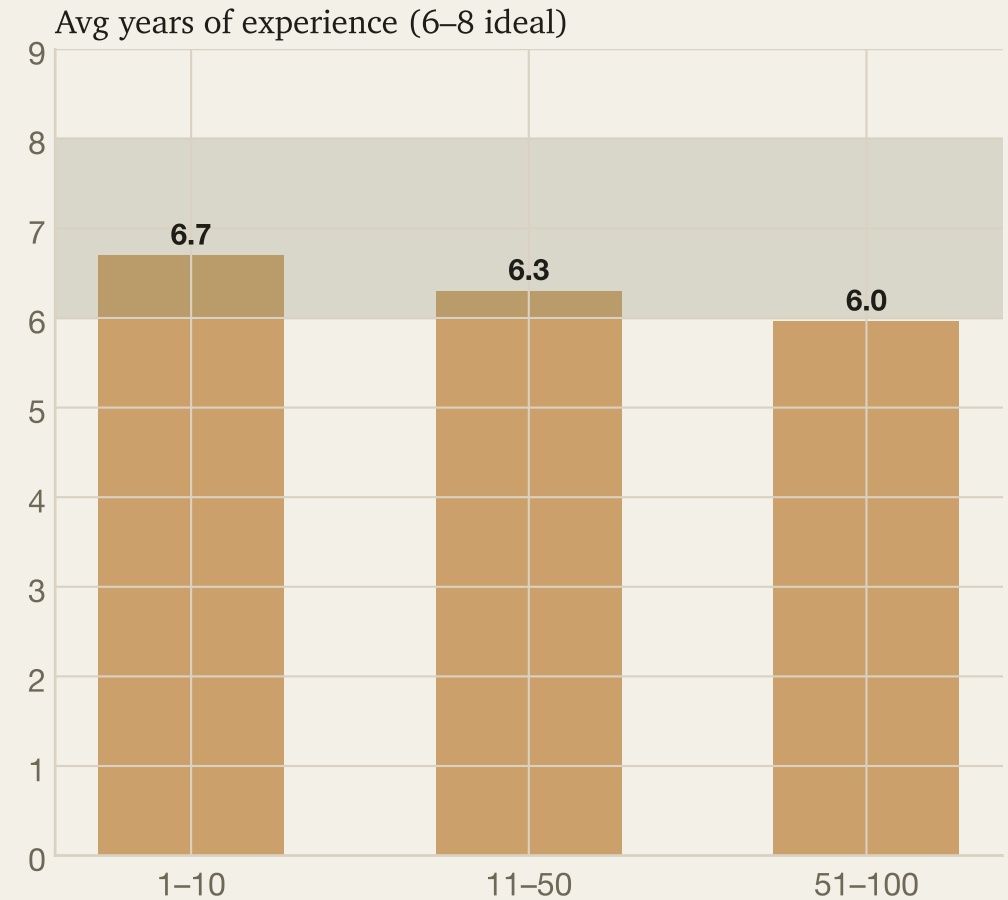
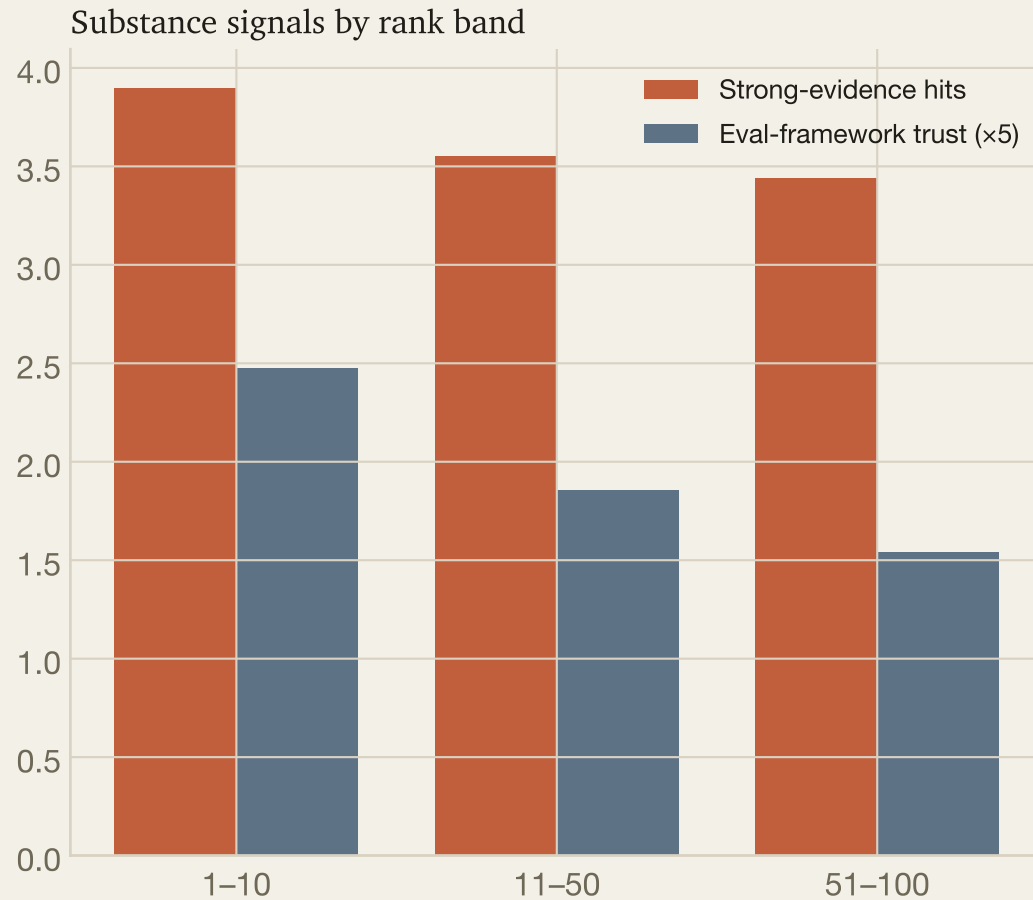
GitHub activity, profile completeness, interview-completion, willingness to relocate.

The ranker separates great from good



A sharper top end lifts NDCG@10 — half the grade. Rank 1 rose to ~0.98, widening the gap to the merely-good.

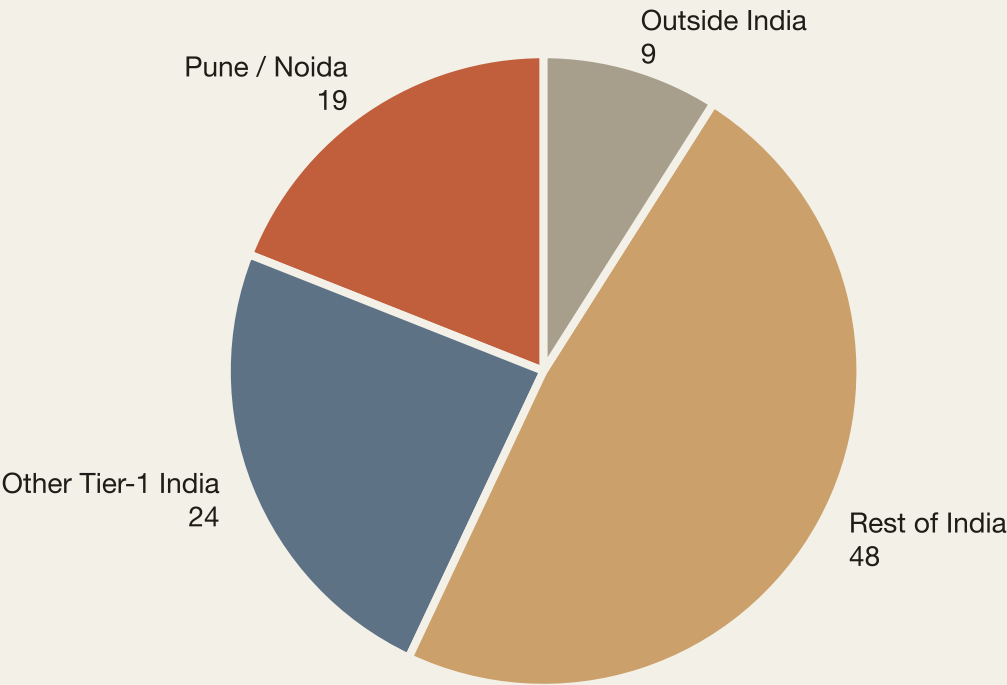
Quality declines smoothly with rank



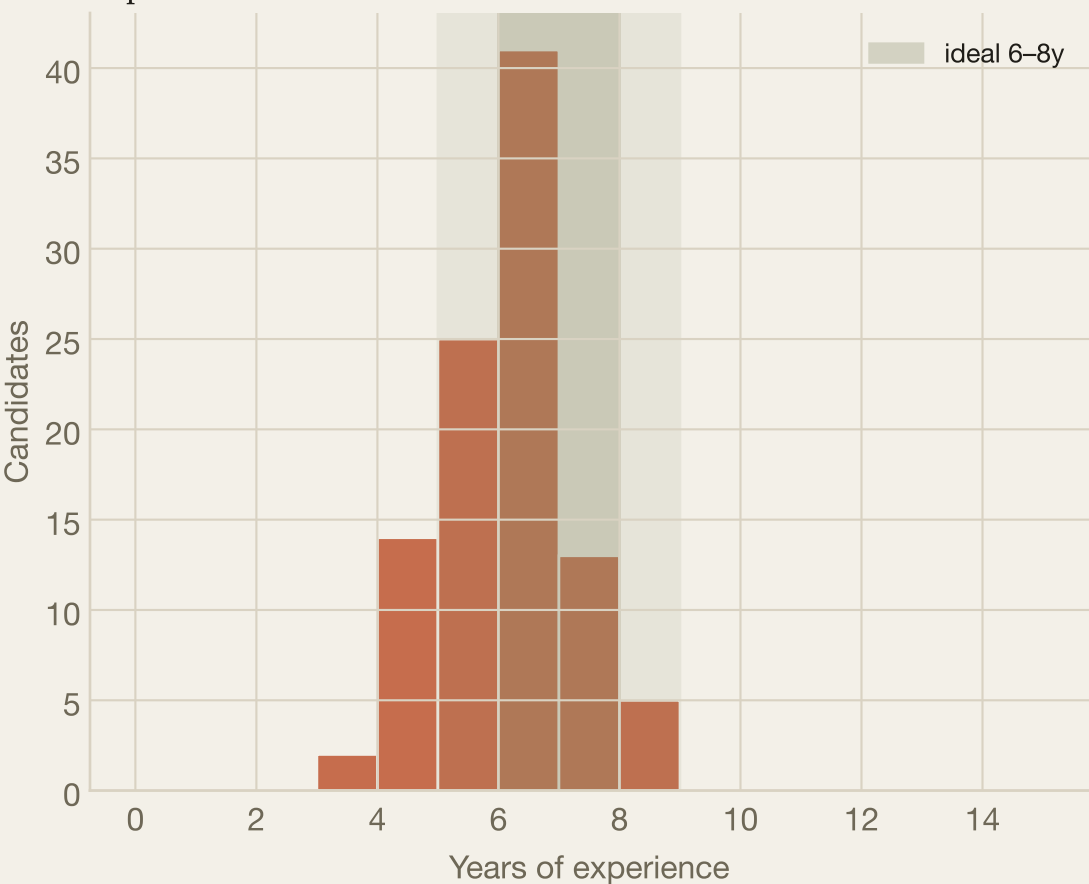
Higher-ranked candidates carry more shipped-system evidence, more rigorous evaluation experience, and sit in the ideal band.

Who makes the top 100

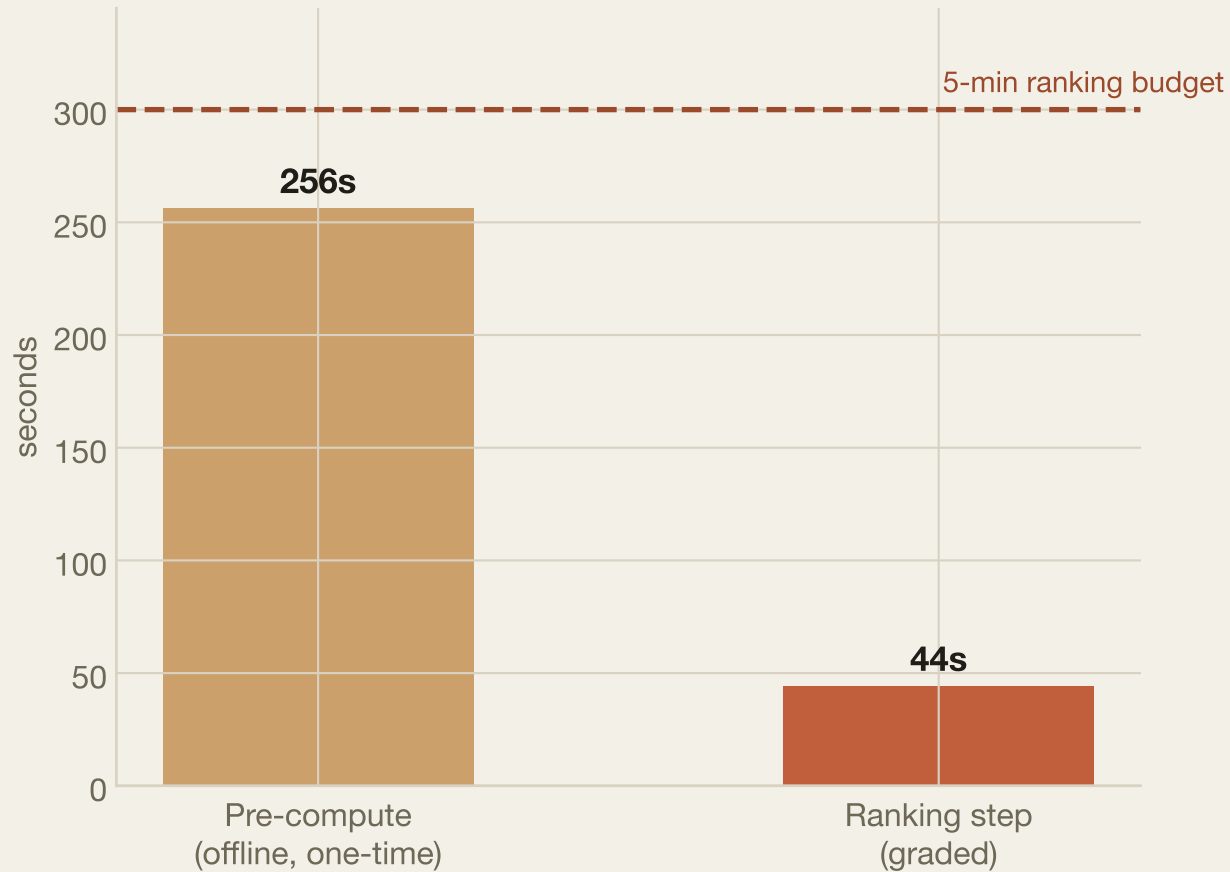
Location mix



Experience distribution



Fast, offline, reproducible



- **CPU-only & offline**

No API calls at ranking time; ~44s on 100K candidates.

- **Pre-compute is separate**

Embeddings (~4.3 min) build via a documented script — outside the budget.

- **Well within limits**

Peak RAM under 16GB; deterministic output on every re-run.

- **Validator-clean**

Passes the bundled `validate_submission.py` unchanged.

Why it wins — and what's next

- **Understands the role, not the keywords**

Rubric + embeddings + free-text evidence working together.

- **Robust to traps**

Honeypots, keyword-stuffers and research-only profiles all sink.

- **Hireability-aware**

Behavioral signals separate "looks great" from "actually available".

- **Explainable & reproducible**

Grounded reasoning per pick; CPU-only, offline, < 5 min, validator-clean.

Next

Cross-encoder re-rank of the top ~200 for finer NDCG@10 · learned component weights once labelled outcomes exist · a richer "product vs services" company graph.